Multiple choice questions

1. A plane glass is placed over a various coloured letters (violet, green, yellow, red) The letter which appears to be raised more is,

(a) red (b) yellow

(c) green (d) violet

2. Two point white dots are 1 mm apart on a black paper. They are viewed by eye of pupil diameter 3 mm approximately. The maximum distance at which these dots can be resolved by the eye is, [take wavelength of light, λ = 500 nm]

(a) 1 m (b) 5 m

(c) 3 m (d) 6m

3. In a Young's double-slit experiment, the slit separation is doubled. To maintain the same fringe spacing on the screen, the screen-to-slit distance D must be changed to,

(a) $2D$ (b) $\frac{D}{2}$

(c) $\sqrt{2}D$ (d) $\frac{D}{\sqrt{2}}$

4. Two coherent monochromatic light beams of intensities I and $4I$ are superposed. The maximum and minimum possible intensities in the resulting beam are [IIT-JEE 1988]

(a) $5I$ and I (b) $5I$ and $3I$

(c) $9I$ and I (d) $9I$ and $3I$

5. When light is incident on a soap film of thickness 5×10^{-5} cm, the wavelength of light reflected maximum in the visible region is 5320 Å. Refractive index of the film will be,

(a) 1.22 (b) 1.33

(c) 1.51 (d) 1.83.

6. First diffraction minimum due to a single slit of width 1.0×10^{-5} cm is at 30°. Then wavelength of light used is,

(a) 400 Å (b) 500 Å

(c) 600 Å (d) 700 Å

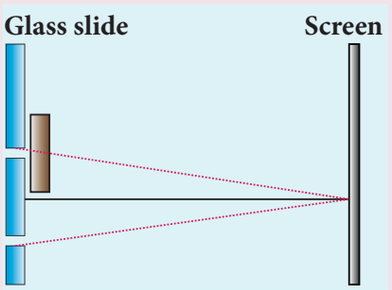
7. A ray of light strikes a glass plate at an angle 60°. If the reflected and refracted rays are perpendicular to each other, the refractive index of the glass is,

(a) $\sqrt{3}$ (b) $\frac{3}{2}$

(c) $\sqrt{\frac{3}{2}}$ (d) 2

8. One of the of Young's double slits is covered with a glass plate as shown in figure. The position of central maximum will,

KF0H6



(a) get shifted downwards

(b) get shifted upwards

(c) will remain the same

(d) data insufficient to conclude

9. Light transmitted by Nicol prism is,

(a) partially polarised

(b) unpolarised

(c) plane polarised

(d) elliptically polarised

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